

Benchmark results compiled with -O2: mean time (microseconds)

Test case	eff	effectful	effective	effective.fstg	effective.lstg	effective.naive	freer	fused	handrolled.list	handrolled.logiet	heftia	mp	mp.safe	mtl.list	mtl.logiet	polysemy
countdown.shallow.10000	281.149	142.74	257.721	2.494	–	86.49	154.12	4.911	–	4.933	172.066	186.394	1100.91	–	4.951	1209.482
countdown.deep.10000	279.041	146.023	162.491	2.494	65.201	168.262	571.096	2683.87	–	4.954	1286.715	191.453	3509.802	–	4.971	2309.474
countdown.very-deep.10000	276.788	162.11	164.138	2.497	65.032	1314.537	5075.962	–	–	4.939	9251.606	490.748	2380.338	–	4.98	11114.513
catch.shallow.10000	400322.2	1065.504	245.976	16.077	–	73.806	35.393	20.195	–	20.36	269.634	835.352	834.281	–	20.257	1117.336
catch.deep.10000	399330.6	1077.138	174.191	16.057	167.612	240.985	34.398	1700.774	–	20.243	3259.689	846.729	851.639	–	20.353	6536.075
catch.very-deep.10000	378063.8	1201.884	175.966	15.978	165.591	1784.356	32.4	–	–	20.39	40653.7	857.593	849.1	–	20.287	40521.762
local.shallow.10000	398140.6	536.764	198.672	2.5	–	134.544	616.257	4.974	–	4.974	918.012	781.972	782.685	–	4.982	2193.356
local.deep.10000	384135.2	537.659	202.792	2.498	201.48	218.587	454.187	4.971	–	4.972	3847.183	790.822	784.5	–	4.98	5099.9
local.very-deep.10000	382941.1	570.613	203.396	2.497	201.966	896.762	456.704	–	–	3524.177	36030.113	789.211	797.438	–	3537.939	28175.638
nondet.shallow.32	24556.725	–	756.505	79.51	–	898.865	3378.8	240.779	347.748	241.682	2006.941	1716.709	1725.028	347.535	225.101	8074.909
nondet.deep.32	50724.2	–	764.697	79.321	990.087	603.822	5710.284	2404.49	62482.6	241.867	5511.312	5635.622	5650.906	62547.275	242.35	8071.544
nondet.very-deep.32	392451.4	–	770.11	79.857	988.955	614.329	25597.575	–	667393.6	248.269	41445.35	38921.3	38913.15	667690.8	248.897	7755.594

Benchmark results compiled with -O2: mean time (% of fastest)

Test case	eff	effectful	effective	effective.fstg	effective.lstg	effective.naive	freer	fused	handrolled.list	handrolled.logiet	heftia	mp	mp.safe	mtl.list	mtl.logiet	polysemy
countdown.shallow.10000	11272.8%	5723.3%	10333.5%	100.0%	–	3467.9%	6179.5%	196.9%	–	197.8%	6899.1%	7473.6%	44141.6%	–	198.5%	48494.9%
countdown.deep.10000	11189.4%	5855.4%	6515.8%	100.0%	2614.5%	6747.2%	22900.6%	107621.7%	–	198.7%	51506.5%	7677.2%	140741.0%	–	199.4%	92608.6%
countdown.very-deep.10000	11083.4%	6491.4%	6572.6%	100.0%	2604.1%	52638.1%	203257.0%	–	–	197.8%	370462.6%	19651.1%	924207.2%	–	199.4%	445059.0%
catch.shallow.10000	2490052.0%	6627.6%	1530.0%	100.0%	–	459.1%	220.2%	125.6%	–	126.6%	1677.2%	5196.0%	5189.3%	–	126.0%	6950.0%
catch.deep.10000	2486944.1%	6708.2%	1084.8%	100.0%	1043.9%	1500.8%	214.2%	10592.1%	–	126.1%	20300.6%	5273.2%	5303.8%	–	126.8%	40705.3%
catch.very-deep.10000	2366191.6%	7522.2%	1101.3%	100.0%	1036.4%	11167.8%	202.8%	–	–	127.6%	254439.7%	5367.4%	5314.3%	–	127.0%	253614.0%
local.shallow.10000	15925725.9%	21470.7%	7946.9%	100.0%	–	5381.8%	24650.4%	198.9%	–	199.0%	36720.7%	31279.1%	31307.6%	–	199.3%	87734.8%
local.deep.10000	15379569.5%	21526.2%	8119.1%	100.0%	8066.6%	8751.5%	18184.2%	199.0%	–	199.0%	154029.1%	31662.0%	31408.9%	–	199.4%	204184.0%
local.very-deep.10000	15337705.7%	22854.4%	8146.5%	100.0%	8089.2%	35917.5%	18292.1%	–	–	141151.7%	1443092.1%	31609.8%	31939.3%	–	141702.9%	1128501.6%
nondet.shallow.32	30885.0%	–	951.5%	100.0%	–	1130.5%	4249.5%	302.8%	437.4%	304.0%	2524.1%	2159.1%	2169.6%	437.1%	283.1%	10155.8%
nondet.deep.32	63948.4%	–	964.1%	100.0%	1248.2%	761.2%	7199.0%	3031.4%	78772.3%	304.9%	6948.2%	7104.9%	7124.1%	78853.8%	305.5%	10175.9%
nondet.very-deep.32	491440.7%	–	964.4%	100.0%	1238.4%	769.3%	32054.1%	–	835732.5%	310.9%	51899.2%	48738.5%	48728.3%	836104.7%	311.7%	9711.8%

Benchmark results compiled with -O2: allocated bytes

Test case	eff	effectful	effective	effective.fstg	effective.lstg	effective.naive	freer	fused	handrolled.list	handrolled.logiet	heftia	mp	mp.safe	mtl.list	mtl.logiet	polysemy
countdown.shallow.10000	2087371	1121570	2480242	24	–	1040121	1600145	24	–	24	1760162	3441170	14241121	–	24	13361481
countdown.deep.10000	2088075	1136505	2000194	24	1200100	3920274	10400173	27121286	–	24	17760483	3442691	41443963	–	24	37363380
countdown.very-deep.10000	2085107	1410353	2000226	104	1200132	20401244	89600730	–	–	104	161763304	3776413	286589490	–	104	253380957
catch.shallow.10000	413467414	7846353	1687795	65585	–	723953	65596	65586	–	65586	4069612	6081394	6081394	–	65586	7764897
catch.deep.10000	414305494	7864171	1527786	65585	1527786	3367875	66058	16165107	–	65585	16870482	6083955	6083955	–	65586	55153744
catch.very-deep.10000	421556572	8163212	1527777	65585	1527787	18074259	70018	–	–	65586	132077892	6139778	6139778	–	65586	371972753
local.shallow.10000	409641299	4326220	1669450	24	–	1029434	4793294	24	–	24	7205058	5513578	5513578	–	24	12630097
local.deep.10000	410402734	4341334	1589474	24	1589474	2549554	4793732	24	–	64	16060654	5515826	5515826	–	64	302322078
local.very-deep.10000	417691622	4648661	1589474	64	1589475	9830346	4797716	–	–	28886111	95215313	5535987	5535986	–	28886124	188651329
nondet.shallow.32	111261132	–	1531330	108345	–	1532765	15889970	3606907	3664101	2030987	13454124	22646075	22646075	3664108	1987330	4429266
nondet.deep.32	383227083	–	1531411	108345	7408482	1535162	45646135	26810555	469819525	2010674	67560404	68636524	68636530	469819536	1969098	45369971
nondet.very-deep.32	4657484945	–	1531410	108345	7408482	1559642	313450797	–	5480085304	2076914	554516648	483091545	483091454	5480086293	2035339	44266212

Benchmark results compiled with -O2: allocated bytes (% of least allocated)

Test case	eff	effectful	effective	effective.fstg	effective.lstg	effective.naive	freer	fused	handrolled.list	handrolled.logiet	heftia	mp	mp.safe	mtl.list	mtl.logiet	polysemy
countdown.shallow.10000	8697379.2%	4673208.3%	1033431.7%	100.0%	–	4333837.5%	6667270.8%	100.0%	–	100.0%	7334008.3%	14338208.3%	59388004.2%	–	100.0%	55672837.5%
countdown.deep.10000	8700312.5%	4735437.5%	8334141.7%	100.0%	5000416.7%	16334475.0%	43334054.2%	11300358.3%	–	100.0%	74002012.5%	14344545.8%	172683179.2%	–	100.0%	155680750.0%
countdown.very-deep.10000	2004910.6%	1356108.7%	1923294.2%	100.0%	1153973.1%	19616580.8%	86154548.1%	–	–	100.0%	155541638.5%	3631166.3%	275566817.3%	–	100.0%	243635535.6%
catch.shallow.10000	630429.8%	11963.6%	2573.4%	100.0%	–	1103.8%	100.0%	100.0%	–	100.0%	6205.1%	9272.5%	9272.5%	–	100.0%	11389.4%
catch.deep.10000	631707.7%	11990.8%	2329.5%	100.0%	2329.5%	5135.1%	100.7%	24647.6%	–	100.0%	25723.1%	9276.4%	9276.4%	–	100.0%	84095.1%
catch.very-deep.10000	642763.7%	12446.8%	2329.5%	100.0%	2329.5%	27558.5%	106.8%	–	–	100.0%	201384.3%	9361.6%	9361.6%	–	100.0%	567161.3%
local.shallow.10000	170638745.8%	1802916.7%	6956041.7%	100.0%	–	4289308.3%	19972958.3%	–	–	100.0%	30021075.0%	22973241.7%	22973241.7%	–	100.0%	52625325.0%
local.deep.10000	171001391.7%	1808891.7%	6622808.3%	100.0%	6622808.3%	10623141.7%	19973883.3%	100.0%	–	–	266.7%	66601891.7%	22982608.3%	–	266.7%	125067904.2%
local.very-deep.10000	652643159.4%	7263532.8%	2483553.1%	100.0%	2483554.7%	15350915.6%	7496431.2%	–	–	45134548.4%	148773926.6%	8649979.7%	8649978.1%	–	45134568.8%	294767717.2%
nondet.shallow.32	102691.5%	–	1413.4%	100.0%	–	1414.7%	14666.1%	3329.1%	3381.9%	1849.6%	12417.9%	20901.8%	20901.8%	3381.9%	1834.3%	40142.7%
nondet.deep.32	353710.0%	–	1413.5%	100.0%	6837.9%	1416.9%	42130.4%	24745.5%	433632.9%	1855.8%	62356.7%	63350.0%	63350.0%	433632.9%	1817.4%	40214.1%
nondet.very-deep.32	4298753.9%	–	1413.5%	100.0%	6837.9%	1439.5%	289308.0%	–	5057995.6%	1916.9%	511806.4%	445882.6%	445882.6%	5057996.5%	1878.6%	40856.7%

Benchmark results compiled with -O0: mean time (microseconds)

Test case	eff	effectful	effective	effective.fstg	effective.lstg	effective.naive	freer	fused	handrolled.list	handrolled.logict	heftia	mp	mp.safe	mtl.list	mtl.logict	polysemy
countdown.shallow.10000	866.258	722.33	1687.371	106.123	–	1685.148	516.948	1703.557	–	565.993	792.759	856.807	2200.959	–	669.582	1748.041
countdown.deep.10000	867.081	727.26	1705.43	106.175	1725.277	3345.491	974.746	10662.95	–	5917.894	2967.139	1475.984	5280.875	–	5208.206	2846.219
countdown.very-deep.10000	910.969	800.642	1737.155	106.078	1713.989	16337.806	5312.859	82674.1	–	48549.575	20043.175	6342.8	28817.138	–	43032.475	11730.944
catch.shallow.10000	371472.4	2122.874	954.964	352.473	–	1070.519	303.176	3598.633	–	222.551	1091.311	1689.797	1695.923	–	253.782	2907.922
catch.deep.10000	368451.6	2144.468	1049.974	345.521	1135.516	3094.384	309.3	15848.487	–	4315.866	7926.322	1676.444	1688.159	–	4129.344	7003.994
catch.very-deep.10000	384279.8	2294.166	1072.698	345.133	1154.688	16576.531	309.35	110292.65	–	45862.85	7745.45	1665.527	1664.378	–	39228.912	40557.675
local.shallow.10000	399788.4	1060.92	637.746	298.428	–	666.043	669.727	1434.17	–	332.067	4766.106	1396.529	1401.706	–	351.334	2718.63
local.deep.10000	384788.7	1075.556	640.509	298.22	646.902	1648.311	673.54	6022.478	–	1638.51	13609.319	1410.093	1407.305	–	1813.544	5695.106
local.very-deep.10000	383264.9	1148.224	650.584	298.846	657.346	8663.422	677.427	38495.825	–	11146.062	8214.9	1424.138	1418.992	–	12243.737	29364.475
nondet.shallow.32	28134.325	–	4197.914	5849.553	–	4307.191	5946.584	2234.161	7003.05	2014.584	7710.5	304.884	3072.256	7542.253	2523.436	13892.144
nondet.deep.32	54159.425	–	4333.808	5888.931	4588.006	9838.269	8000.603	10812.862	81872.65	5284.153	16054.781	7886.603	7831.141	81381.15	6965.269	23349.45
nondet.very-deep.32	392578.2	–	4358.966	5848.25	4622.903	54557.275	28326.275	110878.85	743571.2	74887.45	80595.85	47757.875	48046.225	373098.2	50986.425	99687

Benchmark results compiled with -O0: allocated bytes

Test case	eff	effectful	effective	effective.fstg	effective.lstg	effective.naive	freer	fused	handrolled.listt	handrolled.logict	heftia	mp	mp.safe	mtl.listt	mtl.logict	polysemy
countdown.shallow.10000	6268900	5202083	10801445	400153	–	11601229	6080478	11920971	–	2880389	10640948	7041771	19361627	–	4080275	17281886
countdown.deep.10000	6269552	5218174	11213180	400153	11365142	15371645	14881349	63365432	–	30882667	38643779	9444841	48965635	–	30882914	41284588
countdown.very-deep.10000	6267135	5500291	11319006	400153	11398614	44261452	94088347	524203694	–	282905446	290665289	31071203	315398002	–	272104882	257307421
catch.shallow.10000	415982702	12013517	5434003	1462349	–	6393818	2571789	13075339	–	1593377	7914022	8161829	8161832	–	1673347	22351561
catch.deep.10000	416790307	12032293	5845722	1462350	5837666	10164223	2572605	64806353	–	18046823	33116444	8165128	8164621	–	16926443	57554635
catch.very-deep.10000	424127595	12340561	5951270	1462349	5871213	39053660	2579445	598207170	–	213164916	297290493	8227059	8221509	–	177239336	374381093
local.shallow.10000	411563591	6726770	4008483	1767861	–	4648303	5771851	7688268	–	2327869	14981955	7193877	7193877	–	2327813	14008722
local.deep.10000	412324679	6742834	4020203	1767861	4012086	6818582	5772667	28490777	–	9048648	34185134	7197165	7196661	–	9128590	31611570
local.very-deep.10000	419623357	7058083	4125675	1767861	4045555	21307605	5779507	215712384	–	66655440	227160792	7226325	7220781	–	70335380	190035656
nondet.shallow.32	131921113	–	24814522	53824727	–	29683300	34532261	18238563	69116717	12390869	53827237	30882032	30882027	74123186	17397027	81644041
nondet.deep.32	403888242	–	26178325	53824725	28604571	42408356	64288505	63405695	539108949	39124643	137695708	80929300	80928819	543616138	43632237	200681833
nondet.very-deep.32	4678155598	–	26284095	53824723	28636306	139881902	332096082	469909925	4837677497	280236073	892512168	531355718	531350368	4837187723	279746154	1272018001

Benchmark results compiled with -O0: allocated bytes (% of least allocated)

Test case	eff	effectful	effective	effective.fstg	effective.lstg	effective.naive	freer	fused	handrolled.listt	handrolled.logict	heftia	mp	mp.safe	mtl.listt	mtl.logict	polysemy
countdown.shallow.10000	1566.6%	1300.0%	2699.3%	100.0%	–	2899.2%	1519.5%	2979.1%	–	719.8%	2659.2%	1759.8%	4838.6%	–	1019.7%	4318.8%
countdown.deep.10000	1566.8%	1304.0%	2802.2%	100.0%	2840.2%	3841.4%	3718.9%	15835.3%	–	7717.7%	9657.3%	2360.3%	12236.7%	–	7717.8%	10317.2%
countdown.very-deep.10000	1566.2%	1374.5%	2828.7%	100.0%	2848.6%	11061.1%	23513.1%	131000.8%	–	70699.3%	72638.5%	7764.8%	78819.4%	–	68000.2%	64302.3%
catch.shallow.10000	28446.2%	821.5%	371.6%	100.0%	–	437.2%	175.9%	894.1%	–	109.0%	541.2%	558.1%	558.1%	–	114.4%	1528.5%
catch.deep.10000	28501.4%	822.8%	399.7%	100.0%	399.2%	695.1%	175.9%	4431.7%	–	1234.1%	2264.6%	558.4%	558.3%	–	1157.5%	3935.8%
catch.very-deep.10000	29003.2%	843.9%	407.0%	100.0%	401.5%	2670.6%	176.4%	40907.3%	–	14576.9%	20329.7%	562.6%	562.2%	–	12120.2%	25601.4%
local.shallow.10000	23280.3%	380.5%	226.7%	100.0%	–	262.9%	326.5%	434.9%	–	131.7%	847.5%	406.9%	406.9%	–	131.7%	792.4%
local.deep.10000	23323.4%	381.4%	227.4%	100.0%	226.9%	385.7%	326.5%	1611.6%	–	511.8%	1933.7%	407.1%	407.1%	–	516.4%	1788.1%
local.very-deep.10000	23736.2%	399.2%	233.4%	100.0%	228.8%	1205.3%	326.9%	12201.9%	–	3770.4%	12849.5%	408.8%	408.4%	–	3978.6%	10749.5%
nondet.shallow.32	1064.7%	–	200.3%	434.4%	–	239.6%	278.7%	147.2%	557.8%	100.0%	434.4%	249.2%	249.2%	598.2%	140.4%	658.9%
nondet.deep.32	1542.8%	–	100.0%	205.6%	109.3%	162.0%	245.6%	242.2%	2059.4%	149.5%	526.0%	309.1%	309.1%	2076.6%	166.7%	766.6%
nondet.very-deep.32	17798.4%	–	100.0%	204.8%	108.9%	532.2%	1263.5%	1787.8%	18405.3%	1066.2%	3395.6%	2021.6%	2021.6%	18403.5%	1064.3%	4839.5%